

Cloudflare Radar 2024 Year in Review

Global traffic growth, technology adoption, connectivity disruptions, and security threats observed through Cloudflare's network

17%

YoY global Internet traffic growth

3.3x

Starlink traffic surge

225

major disruptions observed

6.5%

traffic mitigated

Audience: technology executives and network infrastructure leaders

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Global Internet traffic growth year over year

3.3x

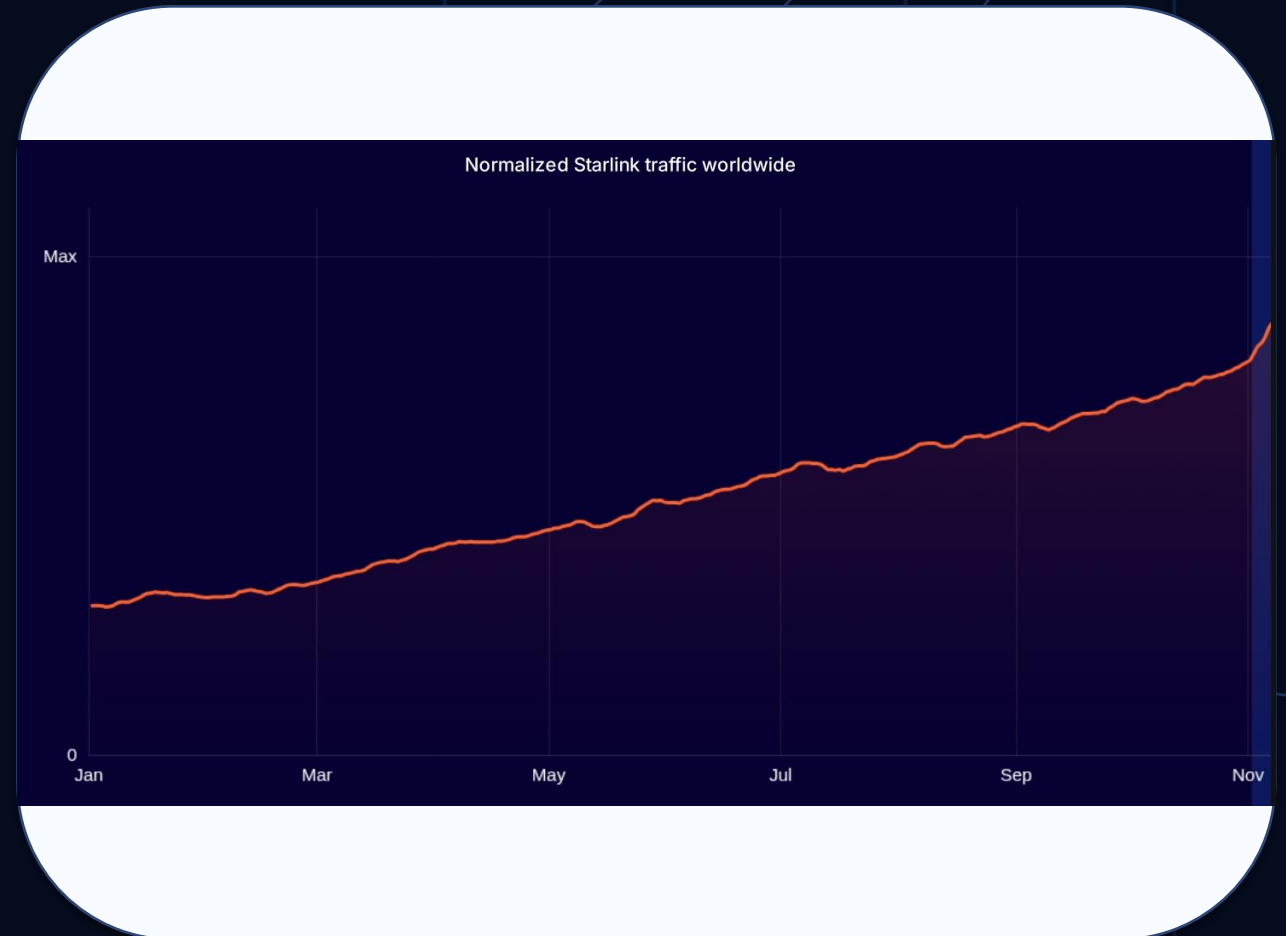
SpaceX Starlink traffic growth in 2024

Executive readout

Internet demand continued to climb as digital services became more embedded in communication, commerce, entertainment, and transportation.

Connectivity signal

Starlink growth points to satellite Internet expanding connectivity into previously underserved locations; trend line is scaled to peak observed request volume.



Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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#1 overall

Google topped the rankings for the second consecutive year.

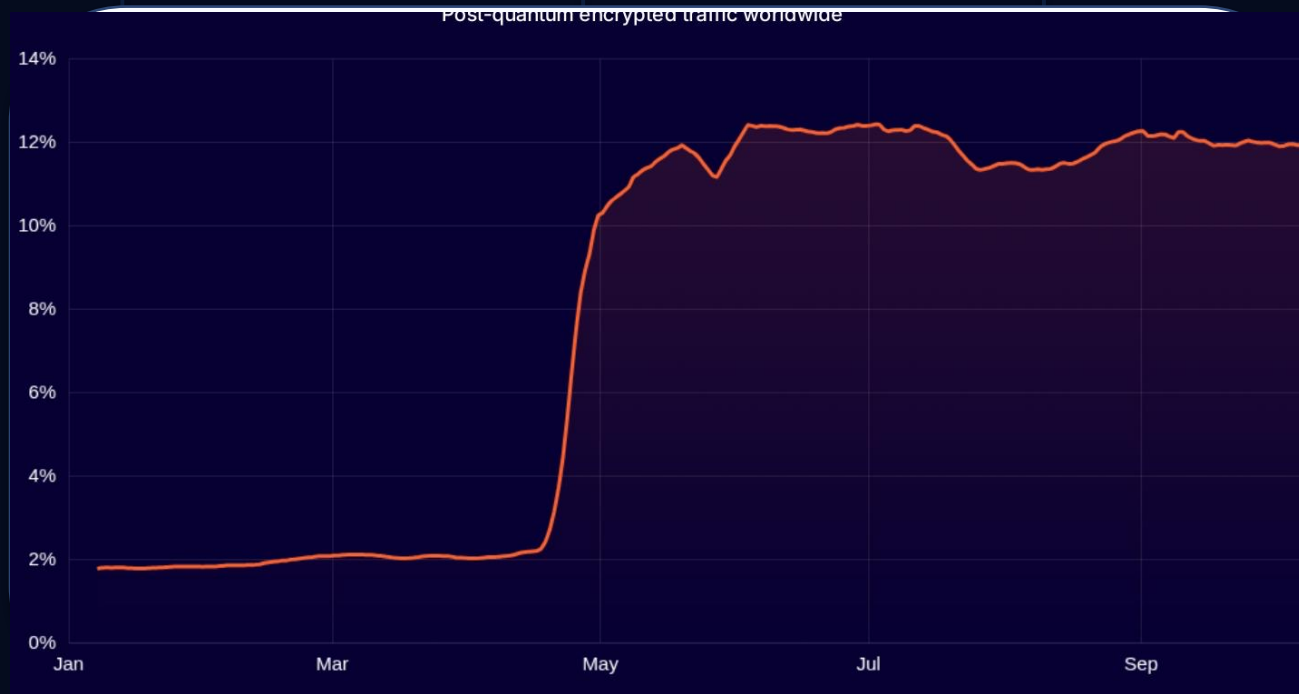
Methodology

Rankings are based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

Market movement

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



2%

Early-2024 post-quantum share of TLS 1.3

12–13%

Rapid climb after Chrome 124 default enablement

21%

Traffic using HTTP/3

~12%

Go share of automated API requests

Trigger event

Google Chrome 124 enabled post-quantum key agreement by default on April 17.

Protocol shift

HTTP/3 runs on QUIC, supporting faster connections, packet-loss mitigation, and encryption by default.

Developer ecosystem

Go became the top API client language, followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

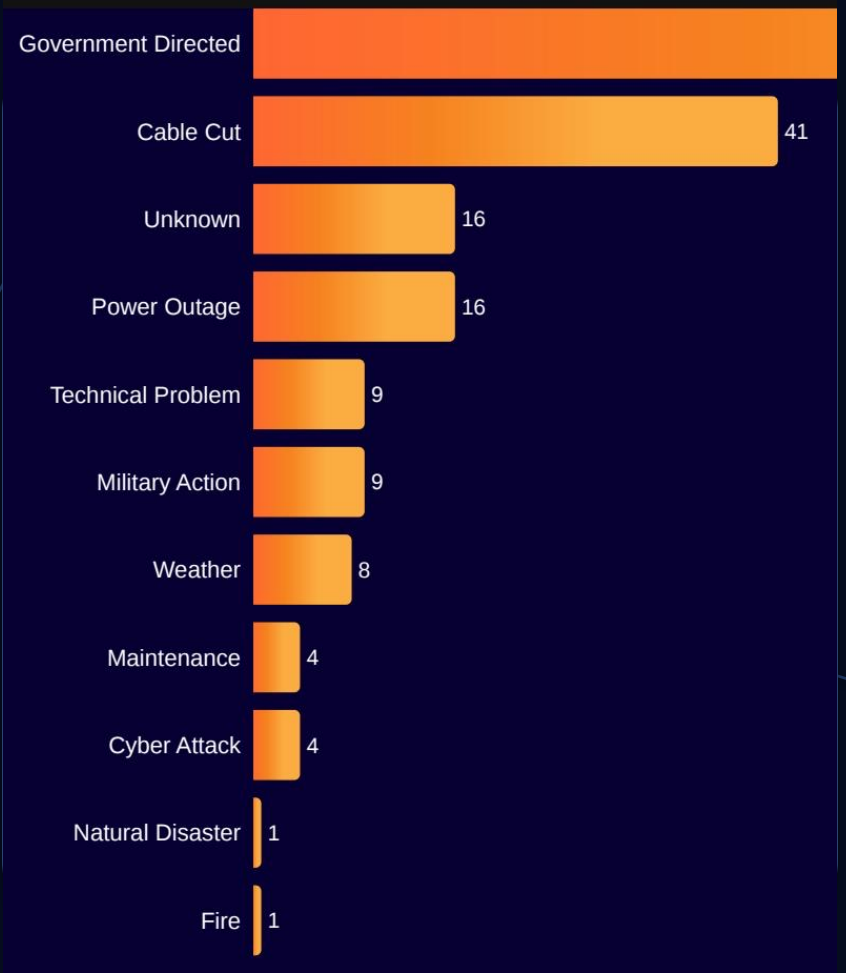
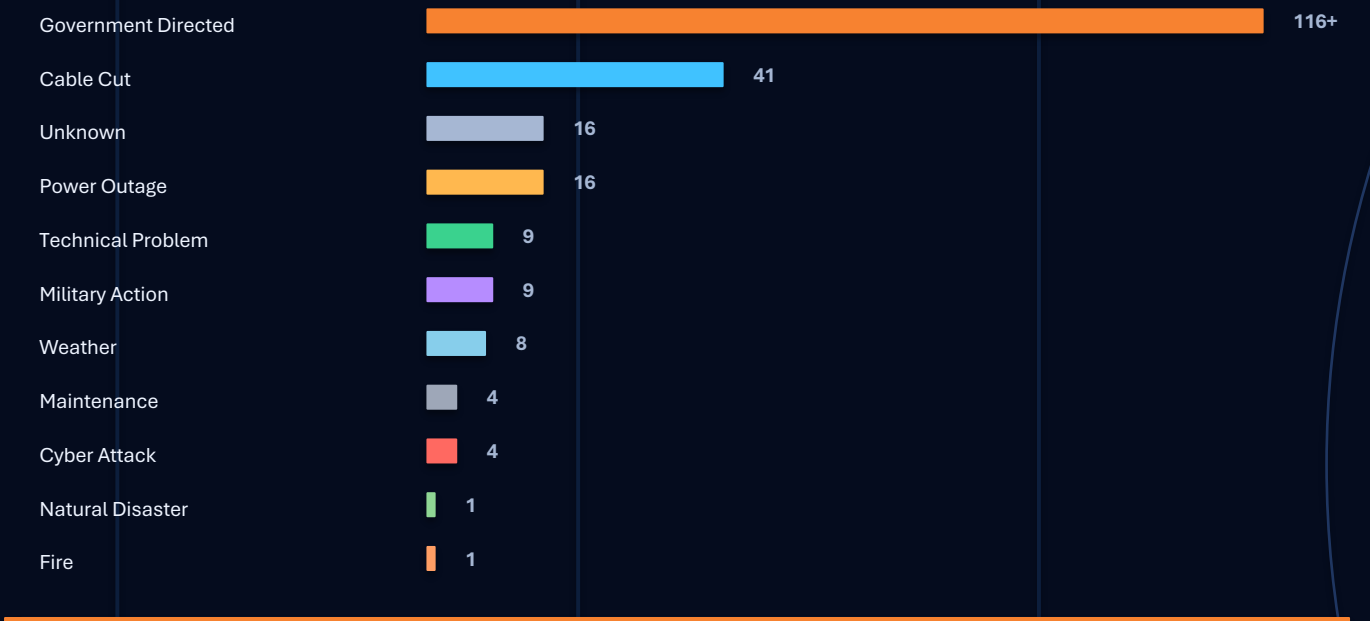
major Internet disruptions globally

116+

government-directed disruptions

41

cable cut incidents



Strategic risk

Government-directed shutdowns were the leading observed disruption cause, exceeding cable cuts by nearly 3x using the reported counts.

Source: Cloudflare Radar 2024 Year in Review

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration



Most-attacked sector

Gambling and Games was the most attacked industry category in Cloudflare's 2024 observations.

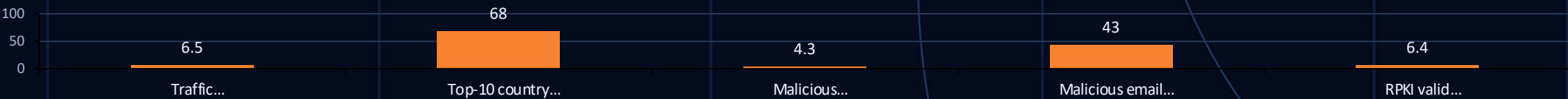
Vulnerability persistence

Log4j (CVE-2021-44228) remains actively targeted more than two years after disclosure.

Routing security

RPKI valid routes increased by 6.4%, improving Internet routing security posture.

Metric



Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic baseline keeps rising

Global Internet traffic grew 17% year over year; capacity plans should assume sustained demand growth.

2

Satellite access is scaling

Starlink traffic surged 3.3x as satellite Internet extended reach into underserved locations.

3

Post-quantum moved from pilot to momentum

TLS 1.3 post-quantum encryption jumped from about 2% to 13% after Chrome 124 default enablement.

4

Connectivity fragility is geopolitical

Government-directed shutdowns led the 225 major disruptions, ahead of cable cuts and power outages.

5

Threat exposure remains persistent

6.5% of traffic was mitigated; Gambling/Games led attacked sectors and Log4j exploitation persisted.

2025 planning lens: design for growth, quantum-safe transition, disruption resilience, and always-on threat mitigation.